

Introductory Econometrics with Recitation: TA Session Week 10

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April 28, 2026

Midterm

- Today we will briefly review the following midterm problems:
 - 4(a), 4(e)
 - 5(a), 5(b)
 - 6(b), 6(e), 6(i)

Midterm Review: 4(a)

- **Question:** The OLS estimator is the sample mean.

Grading Criterion: 4(a)

- **(1 pt)** Correct objective function.
- **(1 pt)** Correct first-order condition.
- **(1 pt)** Correct solution $\hat{\beta} = \bar{Y}$.

Midterm Review: 4(e)

- **Question:** Compare the efficiency of $\hat{\beta}$ and $\tilde{\beta}$.

Grading Criterion: 4(e)

- **(1 pt)** Correct variance of $\hat{\beta}$.
- **(1 pt)** Correct variance of $\tilde{\beta}$.
- **(1 pt)** Correct efficiency conclusion.

Midterm Review: 5(a)

- **Question:** Whether omitting innate ability makes the education coefficient too large or too small.
- Suppose the true model is

$$Wage_i = \beta_0 + \beta_1 Edu_i + \beta_2 Ability_i + v_i,$$

- We make assumption that

$$\beta_2 > 0 \quad \text{and} \quad \text{cov}(Edu_i, Ability_i) > 0,$$

- By the omitted variable bias formula,

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 + \beta_2 \frac{\text{cov}(Edu_i, Ability_i)}{\text{var}(Edu_i)}.$$

- Since the bias term is positive, $\hat{\beta}_1$ will overestimate the true causal effect β_1 .

Grading Criterion: 5(a)

- **(2 pts)** State reasonable assumptions about the omitted variable.
- **(1 pt)** Apply the OVB logic correctly.
- **(1 pt)** Correct direction of bias.

Midterm Review: 5(b)

- **Question:** Give three additional control variables that can reduce omitted variable bias.
- Three useful control variables are:
 - **Labor market experience:** Experience directly affects wages and is related to education decisions.
 - **Family background:** Family background affects educational attainment and also affects later labor-market outcomes through networks, resources, and early investments.
 - **School quality:** Better school quality can raise wages directly through better human capital and is often correlated with how much education a worker obtains.

Grading Criterion: 5(b)

- **(1 pts)** One reasonable control variables.
- **(1 pts)** Reasonable explanation for one control variable.

Midterm Review: 6(b)

- **Question:** Explain why the Bedrooms coefficient barely changes after adding Review Score.
- Coefficient on Bedrooms $72.50 \rightarrow 70.80$ after adding Review Score.
 - Omitted variable bias from leaving out Review Score is small.
- The coefficient on Review Score is **statistically significant**, which suggests that Review Score is correlated with Price.
- Thus, the almost identical coefficient on Bedrooms implies that Review Score is uncorrelated with Bedrooms.

Grading Criterion: 6(b)

- **(2 pts)** State that the coefficient on Review Score is statistically significant.
- **(2 pts)** Explain why Review Score is uncorrelated with Bedrooms.

Midterm Review: 6(e)

- **Question:** Interpret the coefficient on Instant Book in column (3).
- Holding Bedrooms, Bathrooms, Review Score, and Superhost status constant or **other things being equal**, the price of a listing that can be booked without an approval is about \$9.40 higher than an otherwise similar listing that requires an approval.

Grading Criterion: 6(e)

- **(1 pts)** Point out that “hold other things equal”.
- **(2 pts)** Correct interpretation.

Midterm Review: 6(i)

- **Question:** Define heteroskedasticity and explain what heteroskedasticity-robust standard errors.
- Heteroskedasticity:

$$\text{Var}(u_i | X_i)$$

depends on the regressors

- Heteroskedasticity-robust standard errors mean that the reported standard errors are computed using a formula that remains valid even if the error variance is not constant across observations.
 - Still provide valid large-sample inference for confidence intervals and hypothesis tests.

Grading Criterion: 6(i)

- **(2 pts)** Correct explanation of heteroskedasticity.
- **(2 pts)** Correct explanation of robust standard errors.

Quiz 4

- Quiz 4 will be held on May 4.
- It will cover Week 5-8 materials in the TA Session (all slides and practices included).